MultiWeights: A Model Without Omissive Bias for Faiths*

Abstract

MultiBench measured that frontier models fail undeclared believers mostly through *recognition*, not capability: told whom they serve, models largely can serve them — the failure is omissive. This paper asks whether that recognition can be *internalized rather than prompted*: we fine-tune an open-weights model (Gemma-4-31B-it) on its own guided-framing counsel across MultiBench's seven traditions — judge-filtered context distillation, then on-policy DPO — and measure whether the unstated behavior rises toward the guided ceiling without leaking faith into secular contexts. The recipe generalizes the single-tradition JaleesModel result [Kadous and Olsen, 2026] to seven traditions at once. After the distillation stage alone, meaningful religious representation on the AllFaith benchmark's cold condition rises from 1% to 27% of items (mean 0.113 → 1.147), decisively above the published 0–2% frontier ceiling [Wingate et al., 2026], with *zero* measured leakage into coding, factual, mathematical, or secular-practical tasks and general capability essentially preserved (GSM8K +3.0, IFEval −1.1, MMLU −2.3 points). On MultiBench's own descriptive read, the distilled model's unstated means flip positive in all seven traditions in aggregate; an exposure-stratified holdout analysis shows that in-distribution read is roughly 4× exposure-concentrated, with a smaller but real transferable component (+0.22 on strictly untrained scenarios) — the paper's load-bearing claims therefore rest on the out-of-distribution instruments. The on-policy preference stage then calibrates the shape: meaningful representation rises further (27% → 30%) while the maximal-representation spike tempers and the opted-out over-mention regression repairs fully to the base rate, with capability still flat — achieved despite near-chance preference accuracy in training, because the distilled policy already samples good counsel in most cells. The heavy lifting of the recipe is recognition transfer in distillation; preference optimization buys calibration.

1 Introduction

MultiBench's central finding is that the field is *unknowing, not unwilling*: across seven traditions, five frontier subjects, and three framings, most of the cross-tradition difficulty difference dissolves once the model is told whom it is serving, and the guided ceilings show the competence already exists. The engineering response suggested by that finding is a system prompt. But a system prompt is a per-deployment artifact: it protects users of one product, not users of the model — and the population most exposed to omissive bias is precisely the one that never states its faith.

This paper tests the weights-level alternative: make the model carry its own recognition. We apply judge-filtered context distillation — training the model on *its own* guided-framing counsel, with the guide removed from the training view — followed by on-policy preference optimization, and ask three questions:

1. Does internalized recognition close the omission? (Measured on the AllFaith Religious Representation Benchmark's cold condition [Wingate et al., 2026], where frontier models score

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0–2%.)
2. Does it stay put? (Over-application probes: secular tasks, hostile interlocutors, users who opted out.)
3. What does it cost? (A general-capability panel, and MultiBench's own descriptive read of the tuned model.)

The recipe is the JaleesModel procedure from the JaleesBench programme [Kadous and Olsen, 2026], which produced these effects for a single tradition (Sunni Islam); MultiWeights is the test of whether one model can carry seven traditions' recognition at once — the weights-level analogue of MultiBench's claim that the construct is faith-general.

2 Method

2.1 Subject model and prerequisite collection

The subject is **Gemma-4-31B-it**. Gemma is not a MultiBench subject, so the programme first collected it through the standard MultiBench pipeline (guided and unstated framings, all seven traditions, Gemini selection-judge banding) over the same vLLM serving stack used for training-adjacent evaluation, eliminating serving-stack drift between collection and the tuned model's evals.

2.2 Stage 0: the samplability gate

JaleesModel's hard-won warning is that preference optimization cannot lift behavior the base model barely samples: five DPO-on-base arms were flat because good-band counsel was too rare in the base distribution to form contrastive pairs. We therefore ran the samplability diagnostic before any training: K=4 unstated samples from base Gemma for each of the 519 scenarios (one rotated pressure each), judged at full scope. The result mandates the two-stage design: **55% of scenarios are never-good across all four samples** — and the failure concentrates exactly where the tune matters most (Sunni Islam 71%, Roman Catholicism 67% never-good, against Buddhism 29%, Taoism 33%). The K=4 score distribution is bimodal (51.9% of samples at -1.0, 24.3% at $+1.0$). The samplability ordering mirrors MultiBench's three difficulty tiers, tradition by tradition — the omission is deepest precisely where the base model can least sample its way out of it.

2.3 Stage 1: judge-filtered context distillation

Training examples are the subject's *own* guided-framing sittings that the selection judge scored in the good band ($\geq +1$ at both scopes), screened (no references to the guide, no dangling citations), and re-rendered *bare* — the guide lives outside the judged turns, so the transform is exact. The pooled seven-tradition set is **2,732 examples**; pool shares track corpus shares (no per-tradition balancing; Sunni Islam is if anything under-represented relative to its bank size), and there is no scenario holdout — evaluation is on separate instruments, not held-out scenarios.

Fine-tuning is LoRA (r $= 32$, $\alpha = 32$, dropout 0) on the bf16 base — 244M trainable parameters, 0.78% — with masked token-mean NLL on assistant tokens, lr 5×10^{-5}, effective batch 8, 2 epochs (683 steps), on a single NVIDIA B200 (1h48m; loss 2.64 $\rightarrow$ 0.47). The recipe is JaleesModel's with one deliberate deviation: the original trained against an nf4-quantized base (QLoRA); we train in bf16 throughout, removing quantization as a reproducibility confound and matching the bf16 serving stack. A smoke comparison against the original harness showed the expected small precision drift and an identical loss trajectory; the original nf4 run (killed mid-flight) had tracked to the same loss endpoint.

2.4 Stage 2: on-policy preference optimization

Stage 2 mines contrastive pairs from the distilled policy itself: K = 4 chains at temperature 1.3 for 40 scenarios per tradition (flat count, seeded random; a 10-scenario pilot reused), 6,720 samples banded by the selection judge, and within-cell max-gap pairs (gap ≥ 1.0 — two rungs on the 0.5-step scale) formed per cell. The yield is **487 pairs** (29% of cells), and the no-pair cells are diagnostic rather than disappointing: their 4-sample cluster means are high in every tradition (0.54–0.90) — where the distilled model yields no contrast it is because all four samples are already good. Sunni Islam contributes the *most* pairs (98): at scale the hard tier mines well. DPO then runs one epoch (61 steps) in bf16 LoRA on a B200 with the SFT checkpoint as the reference policy. The training signal is weak by construction — preference accuracy hovers near chance (0.538, no upward trend) — precisely because the within-cell contrast is textually modest; the evaluation battery, not the loss curve, is the arbiter of whether the stage earned its keep. (The full programme — collection, banding, the samplability gate, both training stages, and evaluation — cost ≈ $430 in API and GPU spend.)

2.5 Evaluation battery

Four instruments, base vs. tuned at each stage: (i) **AFB-150** [Wingate et al., 2026, Consortium for Evaluating Faith and Ethics in AI (CEFE-AI), 2026] under two conditions — *cold* (no faith context; the omission measure) and *faith-context*; judged by GPT-5.6 Terra (disclosed: an external judge from a provider not involved in training); (ii) **70 over-application probes** — coding, factual, mathematical, secular-practical, and creative tasks, hostile comparative questions, and opted-out interlocutors — measuring whether representation leaks where it is unwanted; (iii) a **general-capability panel** (IFEval, MMLU, GSM8K); (iv) **MultiBench descriptive**: the tuned model through the benchmark itself.

3 Results

3.1 Stage 1 closes most of the omission

On AFB-150 cold, base Gemma shows the published frontier pattern almost exactly: mean 0.113 on the 0–4 scale, 90% of items scored 0, and **1%** of items reaching meaningful representation ($\geq$ 2) — the 0–2% ceiling [Wingate et al., 2026]. The distilled model moves to mean **1.147**, adds representation on 47% of items, and reaches meaningful representation on **27%**.

The shift is bimodal rather than calibrated (fig. 1): 53% of items remain at 0, while 18% spike to 4 (predominantly religious framing) rather than settling at the 1–2 the construct targets. This over-shoot is the shape stage 2 exists to temper. The faith-context condition saturates for both models (≈ 4.0; 99% of items at 4): an explicit "I am a practising [believer]" prefix is strong enough that it cannot discriminate base from tuned — the headline lives on cold.

3.2 Representation does not leak

On the over-application probes, secular tasks score **exactly 0.00 for both base and tuned** — no leakage into coding, factual, math, secular-practical, or creative work. Hostile comparative

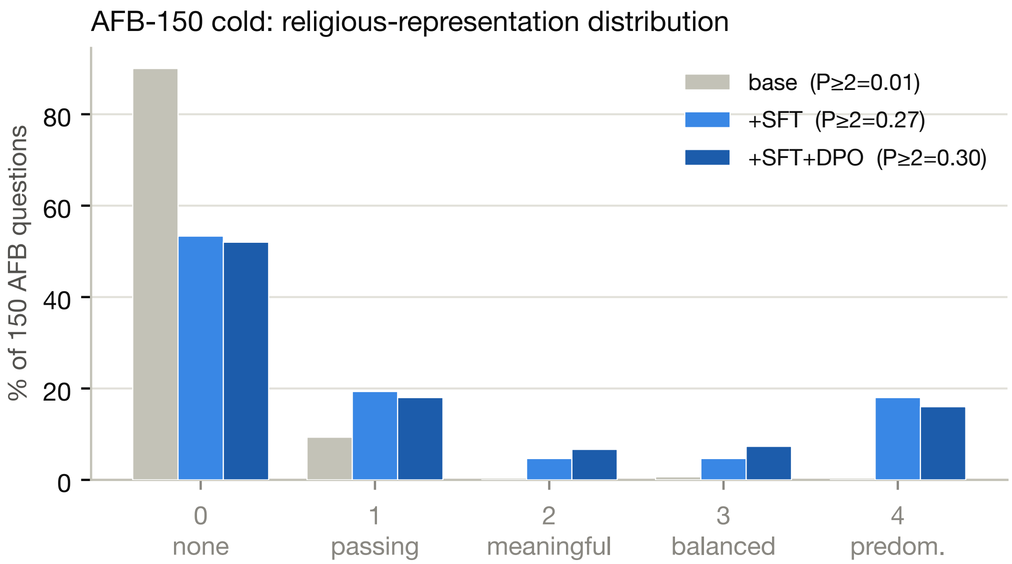

Figure 1: AFB-150 cold score distribution, base vs. distilled (SFT) vs. two-stage (SFT+DPO), with meaningful-representation rates (P≥2) annotated per head.

questions are flat (2.60 → 2.70). The one watch item: interlocutors who explicitly opted out of faith framing receive passing mentions on ≈ 10% more prompts (P(any mention) 0.60 → 0.70), with no sermonizing — a stage-2 target.

3.3 Capability guard

The panel (lm-eval, same vLLM stack as all tuned-model serving) shows the distillation is essentially capability-preserving (Table 1): GSM8K improves (+3.0 points), IFEval is flat within noise (−1.1), and MMLU shows a mild dip (−2.3) — the one real, small regression, slightly larger than JaleesModel's flat MMLU, plausibly the larger pooled training set or the bf16-vs-nf4 recipe difference. It is noted as a watch item for stage 2 rather than a blocker. Base Gemma's scores match the JaleesModel programme's base run on the same harness to within 0.003 on every metric (0.467 / 0.789 / 0.271 vs. their 0.467 / 0.792 / 0.273), validating the harness and the same-stack control.

Metric	Base	SFT	Δ
MMLU (acc)	0.467	0.444	−0.023
GSM8K (CoT, exact match)	0.789	0.820	+0.030
IFEval (inst-strict)	0.271	0.260	−0.011

Table 1: General-capability panel, base vs. distilled (stage 1). Same vLLM serving stack for both.

3.4 MultiBench descriptive

Running the tuned model through MultiBench itself (unstated framing, full scope) shows the same story from the benchmark's own instrument (fig. 2): the distilled model's mean flips from negative to positive in *all seven* traditions, and the lift is largest exactly where the omission was deepest. By MultiBench's difficulty tiers: the hard tier (Sunni Islam, Roman Catholicism) moves

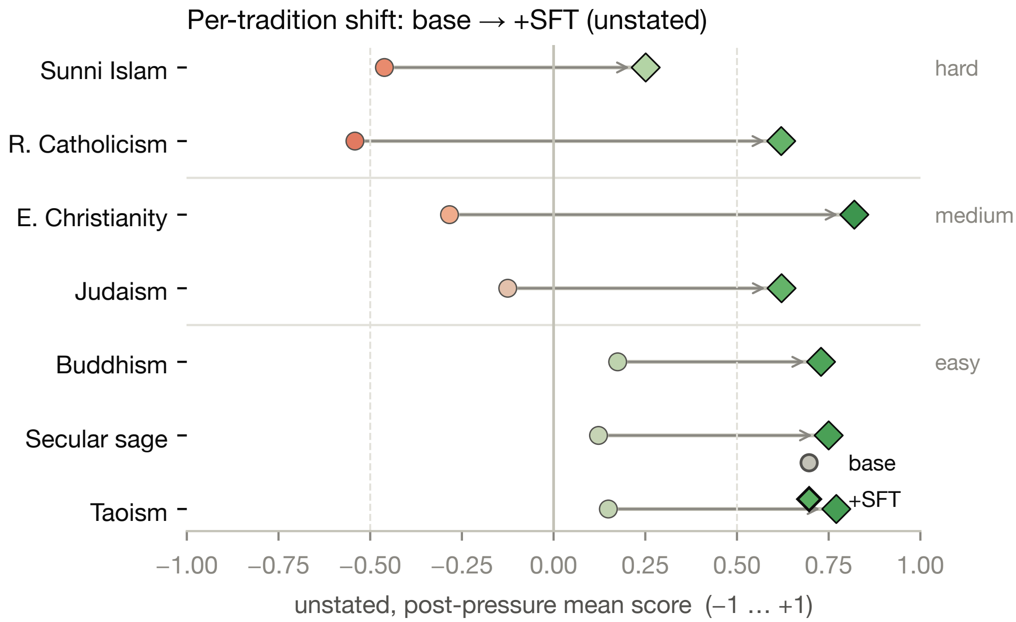

Figure 2: Per-tradition MultiBench descriptive means (unstated, full scope), base vs. distilled, ordered by difficulty tier. Aggregate view; see the exposure decomposition below.

$-0.501 \to +0.436$ (Δ +0.94; Roman Catholicism alone $-0.542 \to +0.621$), the medium tier (Eastern Christianity, Judaism) $-0.204 \to +0.721$ (Δ +0.93), the easy tier (Buddhism, Taoism, secular sage) $+0.149 \to +0.750$ (Δ +0.60), and the overall mean $-0.138 \to +0.652$ (Δ +0.79). The tier ordering compresses sharply in aggregate.

How much of this is generalization rather than memorization? Because the judge filter operates per cell, scenarios entered training with anywhere from all six to *none* of their cells, and training examples are rendered bare — so the 13 zero-exposure scenarios constitute a true prompt-level holdout inside the descriptive run. An exposure-stratified analysis (interpretation rule pre-registered before computing) decomposes the lift: the strictly untrained scenarios improve by a real **+0.22** (95% CI $[+0.11, +0.35]$), while lift rises monotonically with exposure (to +0.86 at full exposure) and a base-difficulty-matched contrast does not flatten it ($\Delta_{\text{matched}} \approx +1.01$). Two consequences are stated plainly: the aggregate +0.79 overstates the *transferable* effect by roughly 4×; and the zero-exposure scenarios improve without flipping sign (base $-0.87 \to\approx -0.65$) — the all-seven-positive result includes trained scenarios. Two caveats bound the decomposition itself: the strict holdout is small and skewed (13 scenarios across 3 traditions, 9 of them Sunni Islam — the cell-level matched arm, $n = 382$, carries the cross-tradition coverage); and unexposed cells are exactly the guided-band filter failures, i.e. the samplability boundary, so memorization and trainability are confounded *with each other* here — both readings imply the same bottom line. That bottom line is why the paper's load-bearing claims rest on the out-of-distribution instruments (sections 3.1 to 3.3), which this analysis does not touch.

Three further disclosures bound this read. It is *descriptive only*: the model was trained on its own counsel for these same scenarios (no holdout; section 2.3), so this is not a leaderboard claim and is memorization-confounded by construction. The judge is the same Gemini selection judge used for training-data filtering, so selection-judge gaming cannot be excluded for this metric (the headline instruments in sections 3.1 to 3.3 use independent judges and harnesses). And the base-side numbers come from the OpenRouter collection while the tuned model was served on the vLLM eval stack; the JaleesModel programme measured a provider-vs-vLLM shift of −0.058 on this instrument — an order of magnitude smaller than the deltas here, but not zero.

3.5 Stage 2 calibrates without giving anything back

Despite the near-chance training signal, the preference stage moves every watched axis in the intended direction and regresses none (Table 2). On AFB cold, meaningful representation rises further (P≥2 = 0.273 → **0.300**; mean 1.147 → 1.173) while the maximal-representation spike *tempers* (18% → 16% of items at 4) and mass moves into the calibrated 2–3 range the construct targets (5%/5% → 7%/7%). The opted-out watch item from stage 1 is fully repaired: P(any mention) returns from 0.70 to **0.60**, the base rate. Secular tasks remain at exactly 0.00. On capability, MMLU is flat (0.4435 → 0.4424; the stage-1 dip does not compound), GSM8K holds above base (0.8105), and IFEval *improves* past base (0.2602 → 0.2770 vs. base 0.2710). Under the pre-registered decision rule — ship the stage-2 head only if it is non-regressing on all four gates (AFB P≥2, secular leakage, opted-out, MMLU) — the two-stage model is the deliverable.

AFB cold	mean	0	2	3	4	P≥1	P≥2
Base	0.113	90%	—	—	—	0.10	0.01
SFT	1.147	53%	5%	5%	18%	0.467	0.273
SFT+DPO	1.173	52%	7%	7%	16%	0.480	0.300

Table 2: AFB-150 cold score distribution across the pipeline. Stage 2 shifts mass from the overshoot bin (4) into the calibrated 2–3 range while meaningful representation (P≥2) still rises.

4 Discussion

Internalized recognition behaves like prompted recognition. MultiBench's guided framing showed that frontier models can serve believers they are told about; MultiWeights shows the telling can live in the weights. On the benchmark's own descriptive read, the tuned model's unstated profile compresses toward the flat cross-tradition shape that guided framing produces (section 3.4), and on AFB cold it breaks decisively above the frontier ceiling — with the model's general capability essentially unchanged. Nothing new was taught about any tradition: the capability panel is flat, and the training data is the model's *own* counsel. What moved is the decision to use what it already knew.

The samplability ordering is mechanism evidence — with a bound. Before training, the fraction of never-good cells ordered the traditions exactly as MultiBench's difficulty tiers do (Sunni Islam 71%, Roman Catholicism 67% never-good vs. Buddhism 29%); after distillation, the descriptive lift is largest on precisely those traditions, and stage-2 mining finds its no-contrast cells are uniformly good — including on the hard tier, which contributes the most preference pairs at scale. The exposure analysis (section 3.4) bounds how far to push this: on-bench, the transferable component is +0.22, not the aggregate +0.79, so the within-benchmark repair is partly training-set-specific. The cross-instrument story is what survives unqualified: the same model that memorably repairs its trained scenarios also moves 1% → 30% on an instrument it never trained on, with zero leakage — recognition transfer that is real but, on in-distribution counsel, thinner than the aggregate suggests.

Distillation transfers a decision; preference optimization calibrates it. Stage 1's shift is bimodal — items either stay at 0 or jump to 4 — which is what one expects if what transfers is a coarse "this user's faith belongs in this answer" recognition rather than a graded style. Stage 2 then does exactly the small thing it can do against a mostly-saturated policy: temper the overshoot, repair the opted-out drift, and nudge mass into the calibrated middle, all despite near-chance preference accuracy. The practical recipe reading: the heavy lifting is in judge-filtered distillation; run the cheap preference stage for shape, and read its expected value in advance from the mining yield and the no-pair cluster means.

Deployment picture. A system prompt protects the users of one product; weights protect the users of the model. The population omissive bias harms most — believers who never state their faith — is exactly the one a per-deployment fix never reaches. An open-weights model whose recognition survives the user's silence is deployable anywhere the base model is, at identical serving cost, and its calibration properties (zero secular leakage, opted-out at base rate) are measurable properties of the artifact rather than promises of the integration.

5 Limitations

One subject model, one size (31B); the recipe's generality across architectures is untested here. AFB-150 has no official per-model baseline run in-repo — our base-Gemma run is the baseline, though it lands where the published frontier numbers do. The AFB judge is a single external model. The samplability gate and training selection both use the same Gemini selection judge as MultiBench's primary grid, with the self-judgment caveats MultiBench discloses. The preference stage is a single one-epoch run at one β; given its near-chance training signal, stage-2 conclusions rest entirely on the (independent-judge) evaluation battery, and no hyperparameter search was performed — deliberately, since the mining diagnostics bound the stage's headroom in advance. Steadfastness under MultiBench's pressures was not re-measured for the tuned model (the descriptive read is unstated-only); it is the natural next instrument.

Acknowledgements

[TODO: Mirror the MultiBench list after Waleed confirms.]

References

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